Experiment-1

Name: - J pranitha **Regno: -**192472073

Course Code: -MLA0305 **Slot-**B

Title: -Installation and Configuration of Anaconda Navigator, Python, TensorFlow, Keras and Gymnasium

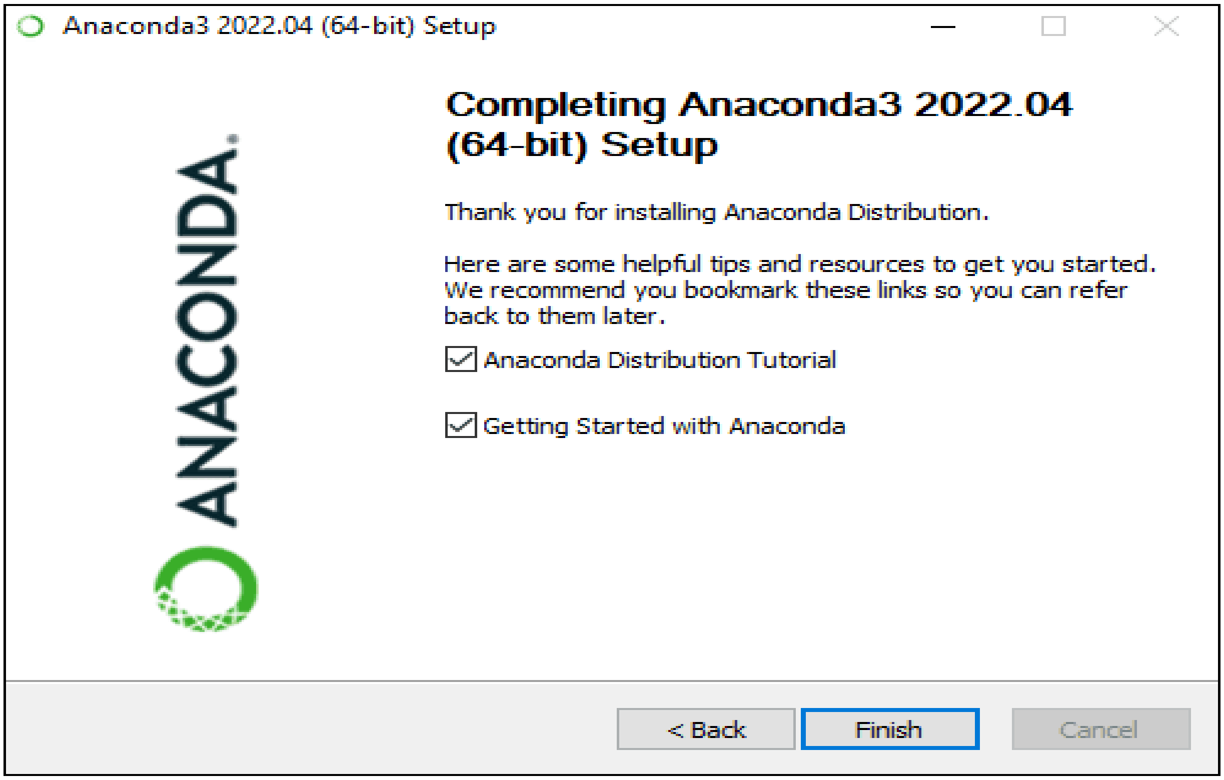
AIM

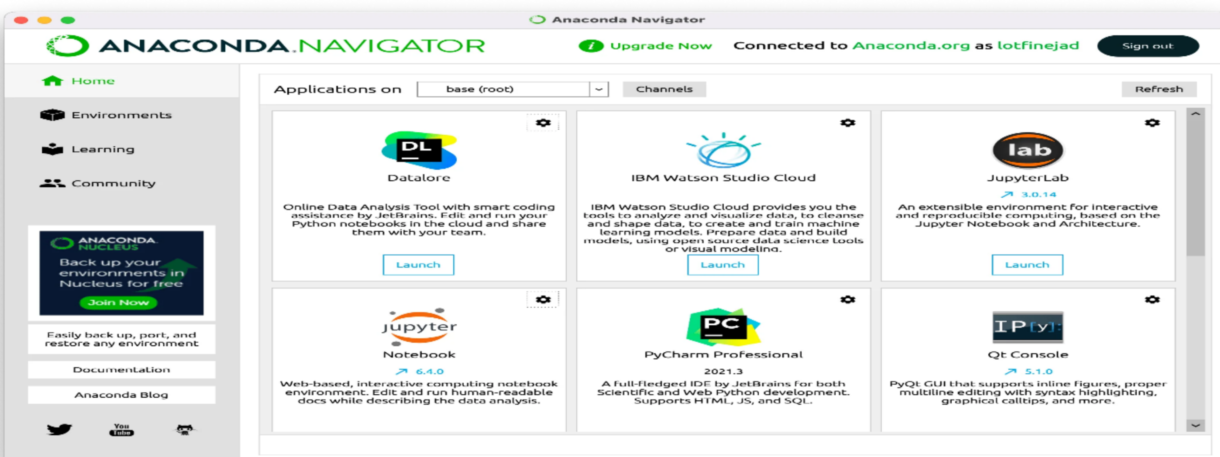
To install and configure Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium, and to verify that all the required tools and libraries are working correctly for Reinforcement Learning experiments.

PROCEDURE

1. Install Anaconda Distribution on the computer.
2. Open Anaconda Prompt or Anaconda Navigator.
3. Create a separate environment for Reinforcement Learning.
4. Activate the newly created environment.
5. Verify the installed Python version.
6. Install TensorFlow using pip.
7. Install Keras and verify its installation.
8. Install Gymnasium, which provides standard environments for Reinforcement Learning experiments.
9. Install supporting libraries such as NumPy, Pandas, and Matplotlib.
10. Open Jupyter Notebook or Google Colab for writing and executing Python programs.
11. Import all the installed libraries.
12. Check and display their versions to verify successful installation.
13. Create a simple Gymnasium environment such as CartPole.
14. Reset the environment and verify that the environment returns a valid initial state.
15. Execute a few random actions to confirm that the Reinforcement Learning environment is functioning correctly.
16. Record the installation results and screenshots of the successful execution.

VISUALIZATIONS





Summary Table

S.No.	Software / Library	Purpose	Verification
1	Anaconda	Python distribution and environment management	Anaconda Prompt opened successfully
2	Python	Programming language for Reinforcement Learning	Python version displayed
3	TensorFlow	Machine Learning and Deep Learning framework	TensorFlow imported successfully
4	Keras	High-level Deep Learning API	Keras imported successfully
5	Gymnasium	Reinforcement Learning environment framework	Gymnasium environment created successfully
6	NumPy	Numerical and array operations	NumPy imported successfully
7	Pandas	Data handling and analysis	Pandas imported successfully
8	Matplotlib	Data visualization and plotting	Graph generated successfully
9	CartPole Environment	Sample RL environment for testing	Environment reset and actions executed
10	Jupyter Notebook / Google Colab	Code execution environment	Python code executed successfully

Result

The required Reinforcement Learning software and libraries are successfully installed and configured, and a Gymnasium environment executes successfully without errors